4.4 The Cosine Function and Inverse Cosine Function

The cosine function is a function with $\mathbb{R}$ as its domain and $[-1, 1]$ as its range. We write $y = \cos x$ and $y = \cos^{-1} x$ or $y = \arccos(x)$ to represent the cosine function and the inverse cosine function, respectively. Since $\cos(x + 2\pi) = \cos x$ is true for all real numbers x and $\cos(x + p)$ need not be equal to $\cos x$ for $0 < p < 2\pi$, $x \in \mathbb{R}$, the period of $y = \cos x$ is 2π.

4.4.1 Graph of cosine function

The graph of cosine function is the graph of $y = \cos x$, where x is a real number. Since cosine function is of period 2π, the graph of cosine function is repeating the same pattern in each of the intervals $\cdots, [-4\pi, -2\pi], [-2\pi, 0], [0, 2\pi], [2\pi, 4\pi], [4\pi, 6\pi], \cdots$. Therefore, it suffices to determine the portion of the graph of cosine function for $x \in [0, 2\pi]$. We construct the following table to identify some known coordinate pairs (x, y) for points on the graph of $y = \cos x$, $x \in [0, 2\pi]$.

x (in radian)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1

The table shows that the graph of $y = \cos x$, $0 \le x \le 2\pi$, begins at (0,1). As x increases from 0 to π, the value of $y = \cos x$ decreases from 1 to -1. As x increases from π to 2π, the value of y increases from -1 to 1. Plot the points listed in the table and connect them with a smooth curve. The portion of the graph is shown in Fig. 4.10.

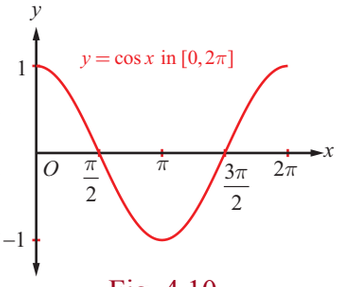


Fig. 4.10

The **graph of** $y = \cos x$, $x \in \mathbb{R}$ consists of repetitions of the above portion on either side of the interval $[0, 2\pi]$ and is shown in Fig. 4.11.

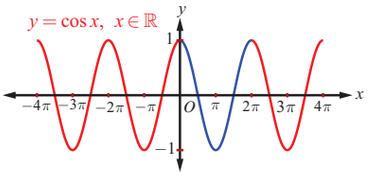


Fig. 4.11

From the graph of cosine function, observe that $\cos x$ is positive in the first quadrant $\left(\text{for } 0 \le x \le \frac{\pi}{2}\right)$, negative in the second quadrant $\left(\text{for } \frac{\pi}{2} < x \le \pi\right)$ and third quadrant $\left(\text{for } \pi < x < \frac{3\pi}{2}\right)$ and again it is positive in the fourth quadrant $\left(\text{for } \frac{3\pi}{2} < x < 2\pi\right)$.

Note

We see from the graph that $\cos(-x) = \cos x$ for all x, which asserts that $y = \cos x$ is an even function.

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